

Supporting Information for:

Child-Directed Speech Is Semantically Constrained: Evidence from Metaphoricity and Concreteness

Note. The analyses reported here are supplementary and descriptive. They are included to enhance transparency and provide broader developmental context but do not form part of the manuscript’s central argument.

A. Examples of MelBERT Annotations of Metaphorical Word Use

The following examples illustrate MelBERT annotations of words used in a nonliteral sense. Metaphorical words in text are tagged with a hashtag symbol.

- *“#that must have #taken a #long time”*
“That” is annotated as metaphorical because it refers to an event rather than to an entity physically present in the speech situation. “Take” is used in a nonliteral sense because no physical grasping is taking place; instead, it reflects a metaphorical mapping between taking an object and taking time. Similarly, “long” is not used in its primary sense as an object’s physical property, but maps from the domain of space to the domain of time. In the same vein, “long” is not employed in its primary sense of an object’s physical property and rather maps from the domain of space to the domain of time.
- *“You could #approach your writing #from #that #angle”.*
The direct meaning of “approach” refers to moving nearer to an object in space; here, it instead expresses a way of dealing with a subject, mapping from physical movement to the mental domain of writing. “From that angle” metaphorically maps from the geometric dimension to an imaginative act of adhering to a specific idea or point of view, and thus implies adopting a different perspective on the writing task.
- *“He does not #see formal #codes of practice as the real outcome”.*
In this sentence, “see” does not refer to vision, but rather to understanding or subjective evaluation. Likewise, the primary literal definition for “code” (a system of figures or symbols in need of deciphering) clashes with its usage as a set of accepted (sometimes, implicit) rules for the society to follow. Metaphorical mapping emerges here to connect the source domain of secretive manipulation of symbols to the abstract target of best practices in human interaction.
- *“#go #back #to sleep”*
In its primary sense, “go” refers to physical movement, but here it is used to encourage the child to return to a previous state, namely sleep. The meaning of both “back” and “to” is metaphorical here, since “back” is mapped from the space to time dimension, and “to” – from the domain of physical direction to the desired end state of mind.

B. MelBERT Annotations Diverging from Human Judgments

Table S1 presents a representative selection of MelBERT annotations that human annotators judged to be counterintuitive or incorrect. The examples illustrate recurring patterns in which the model diverged from human metaphor judgments.

Table S1: Examples of MelBERT Annotation Errors

Source File	Speech Genre	Annotated Example	Comments	Error Type
Anne 011007	CDS	are you #making a big mess - #big	The word “big” is annotated only on its second occurrence, although “mess” is an abstract concept that cannot physically be big or small, so it would also be appropriate to label its first occurrence as metaphorical.	Labeling inconsistency within an utterance
Anne 011007	CDS	are you going to help me do #these puzzles	The way MelBERT was trained, it tends to over-annotate demonstrative pronouns. However, in cases where it is clear from the context that the demonstrative use is attributed to the tangible things in front of the speaker – like the puzzles in this utterance – it is not metaphorical and thus should not be labeled.	Over-annotation of demonstratives
Anne 011007	CDS	#that baby’s not crying	As in the previous example, the baby (or a doll, as might also be the case in child-directed speech) is directly present or contextually identifiable, and need not be labeled as metaphorical	Over-annotation of demonstratives
Anne 011007	CDS	is it pants or a #hat	Here, the basic meaning of the lexical unit “hat” corresponds to the contextual meaning of a physical piece of clothing (in line with “pants”), and the metaphorical annotation is not necessary.	Misattribution of the metaphorical word sense
Becky 021115	CDS	he’s a #ginger #cat like Sukie isn’t he	The basic meaning of the word “cat” in this utterance is a direct observation or description of the animal. Yet, in this utterance it is annotated as metaphorical along with “ginger”.	Misattribution of the metaphorical word sense

Table S1: Examples of MelBERT Annotation Errors

Source File	Speech Genre	Annotated Example	Comments	Error Type
Dominic 020328	CDS	I'm going to get it to #attack you	In this sentence, the contextual meaning of the word "get" is different from its basic meaning of receiving or obtaining but rather is a metaphorical manifestation of the meaning equivalent of "forcing" someone. Thus, it should be labeled as such.	Omission of the metaphorical sense
Eleanor 030006a	CDS	was #that on television	The meaning of the preposition "on" is not equivalent to its basic meaning of "on top of" or "being supported by the surface". While usually annotated as metaphorical in this prepositional phrase, the labeling of "on" is sometimes omitted, as in this case.	Omission of the metaphorical sense
HM6 119	ADS	well there are a number of, of #points there and I'll, I'll try and cover them all.	Clearly, the verb "cover" in this context is metaphorical. The basic meaning of "forming a layer over something in order to hide or protect it" is incongruent to the sense used in the utterance.	Omission of the metaphorical sense
JJH 43	ADS	the #strengths and weaknesses that some people may #have #in your #team.	In this case, both "strengths" and "weaknesses" are used metaphorically rather than in reference to physical abilities or their absence. Nevertheless, while "strengths" is correctly annotated, "weaknesses" misses its labeling.	Labeling inconsistency within an utterance

Table S1: Examples of MelBERT Annotation Errors

Source File	Speech Genre	Annotated Example	Comments	Error Type
H4A 13	ADS	Incidentally, I can't be with you because I'm trying to ensure that #in the #next general election Labour will win more seats.	The contextual meaning of "seats" in this example diverges from the literal meaning of a place or object used for sitting. Therefore, it requires a metaphorical annotation.	Omission of the metaphorical sense
KB8 4093	ADS	when I saw #that woman crossing the road I thought it was her	The woman in this utterance refers to the direct experience of the speaker and is identified with a literal usage of the demonstrative. In this particular case, "that" corresponds to the basic meaning of reference to the person and should not be labeled as metaphorical.	Over-annotation of demonstratives
CS2 12	Written	He may prefer only to claim that they #apply once ethical language and judgement	The word "language" should be annotated as metaphorical here because its contextual meaning contrasts with the basic sense of a communication system, and rather specifies a certain style of speaking or writing.	Omission of the metaphorical sense
A0U 714	Written	She #sees: not Harriet but her mother.	Unlike the metaphorical usage of "I see" to mean "I understand", here seeing refers to the direct observation with one's eyes. In this utterance, the basic meaning is preserved and should not be labeled as metaphorical.	Misattribution of the metaphorical word sense

Table S1: Examples of MelBERT Annotation Errors

Source File	Speech Genre	Annotated Example	Comments	Error Type
EF1 46	Written	He was thirty eight years old with a youthfully handsome face ... muscular physique which he #kept #in shape #with a daily five kilometre run	Here, “with” is annotated in the second case but not the first one. Arguably, in the first instance, “with” is also used metaphorically, since its meaning in the utterance deviates from the basic dictionary definition of “being together in the same place”.	Labeling inconsistency within an utterance

Across these cases, MelBERT’s errors can be grouped into four recurring categories:

- **Omission of metaphorical senses:** Words judged metaphorical by human annotators are not marked by the model.
- **Misclassification of literal meanings:** Lexical items used in their literal sense are incorrectly labeled as metaphorical.
- **Inconsistent labeling:** The same word or semantically related words are labeled inconsistently, sometimes within a single utterance.
- **Overannotation of demonstratives:** Demonstratives such as *this*, *that*, *these*, and *those* are frequently labeled as metaphorical even when they refer to physically present entities.

Overall, these examples suggest that many MelBERT errors reflect the same ambiguity that makes metaphor annotation challenging for human annotators.

C. Lexical Diversity: Comparison Across Genres

As a supplementary baseline analysis, we also examined lexical diversity across registers. Reduced lexical diversity and high redundancy in CDS are well established findings in the literature (Genovese, 2020; Mintz et al., 2002; Newport et al., 1977; Schwab & Lew-Williams, 2016; Snow, 1972). Caregiver speech has repeatedly been shown to rely on restricted vocabularies, recurrent lexical frames, and structured repetition patterns that may facilitate segmentation and early lexical learning (Goldfield, 1999; Lester et al., 2022).

Lexical diversity (LD) refers to the range and variety of vocabulary used in a text. Often used interchangeably with the terms lexical variation or lexical richness, LD is a phenomenon

measured through a wide variety of indices (**McCarthy2007Vocd:Evaluation**). The most common measure is the type–token ratio (TTR). However, this measure is significantly affected by sample size, and small samples can appear highly diverse simply because they contain only a few tokens. In longer samples, speakers must rely on repeated use of closed-class grammatical or function words, such as articles, prepositions, and auxiliary or modal verbs, which systematically decreases lexical diversity ratios (Yang, 2022).

Over the past decades, other computational measures have been introduced as promising alternatives. In particular, we focus on two methods for estimating lexical diversity in our corpora: voc-D (MacWhinney, 2000) and MATTR (moving average TTR) (Covington & McFall, 2010). The voc-D algorithm outputs an LD index calculated through a series of TTR samplings and curve fittings (**McCarthy2007Vocd:Evaluation**). In one study of young children’s expressive language, only the voc-D measure met the dual challenges of capturing a clear developmental growth profile across the age range and distinguishing between typically developing children and those with known expressive language impairments (Yang, 2022).

MATTR takes a different approach to minimizing text-length effects by calculating a moving-average TTR over all segments of a given length. For a segment length of, for example, 50 tokens, TTR is computed for tokens 1–50, then 2–51, 3–52, and so on, and the resulting values are averaged to produce the final MATTR score (Covington & McFall, 2010; Zenker & Critten, 2021). MATTR has been shown to be resistant to text-length effects and is considered one of the most stable lexical diversity measures (Zenker & Critten, 2021). A segment length (window size) of 50 tokens was used for the moving average calculations for all corpora.

To compare lexical diversity across both measures, we normalized the output values. Normalization is essential when numerical values are expressed on different scales (Peña, 2022). Specifically, we rescaled the features to a range of $[0, 1]$, a special case of min-max scaling (Subasi, 2020). In addition, we collected the 20 most frequent metaphorical word types in each language register and estimated their percentage share among all metaphors within the corpus. Using both measures provides a broader descriptive profile of lexical diversity across registers. All calculations were performed in Python (Van Rossum & Drake, 2009) using the “LexicalRichness” package (Shen, 2022).

Apart from the CDS, ADS, and written registers, we also estimated lexical diversity in the corresponding child-speech corpus. This allowed us to compare child speech measurements directly with their CDS equivalents.

To compare lexical diversity across registers, we conducted a one-way ANOVA on session-level MATTR scores for CDS, ADS, and written language. To examine whether lexical diversity in CDS changed with child age, a linear mixed-effects model was fitted. The dependent variable was the session-level MATTR index within the CDS corpus. Target child age served as the fixed effect baseline, with random intercepts applied across the 14 localized caregiver–child dyads to structure group-level variability:

$$\text{LexDiv_CDS}_{ij} = \beta_0 + \beta_1 \text{Age}_{ij} + u_{0j} + \epsilon_{ij}$$

where:

- LexDiv_CDS_{ij} represents the lexical diversity index of CDS in session i for dyad j ,

- β_0 is the fixed intercept,
- β_1 is the fixed slope associated with child age,
- u_{0j} is the random intercept for caregiver–child dyad j ,
- ϵ_{ij} is the residual error term.

Random slopes for age were excluded because of the small number of caregiver–child dyads ($n = 14$) and the risk of model convergence problems. Table S2 summarizes the lexical diversity results across language genres: it is lowest in CDS and highest in written language, both for general and metaphorical vocabulary. These findings suggest that written texts exhibit substantially greater lexical richness than both CDS and ADS, consistent with their broader communicative and stylistic functions. In contrast, the lower diversity observed in CDS reflects its simplified and repetitive structure, which is tailored to the needs of young language learners. This pattern is consistent with the descriptive expectation that CDS is lexically less diverse than adult-directed and written registers.

Table S2: Lexical diversity measures (MATTR and voc-D) for all words and metaphorical words across child speech, CDS, spoken, and written genres.

Measure	CS	CDS	ADS	Written
All words				
MATTR	0.56	0.67	0.74	0.81
voc-D	78	85	122	217
Metaphorical words				
MATTR	0.36	0.42	0.57	0.71
voc-D	13.5	13.3	38	92.3

We analyzed lexical diversity across registers using MATTR scores computed per session (window size: 50 tokens). A one-way ANOVA revealed a highly significant effect of register on lexical diversity ($F(2, 980) = 1249.26$, $p < .001$). Tukey’s HSD test indicated significant pairwise differences between all registers ($p < .001$) (see Table S3). Mean MATTR scores followed the expected pattern of increasing lexical diversity from CDS to ADS to written language (CDS: 0.673; ADS: 0.739; written: 0.813).

In Figure S1, the normalized values of lexical diversity are shown for both general vocabulary (all words) and metaphorical vocabulary. Across all measures, child-directed speech exhibits lower lexical richness than the other language registers. The differences between registers are pronounced for both overall vocabulary and metaphorical vocabulary, with CDS showing consistently lower diversity in metaphorically used words compared to ADS and written texts. This pattern suggests a relative linguistic scarcity and higher re-use of non-literal expressions in child-directed speech. The corresponding values for MATTR and voc-D, computed for all words and for metaphorical words only, are reported in Table S4.

Table S3: Post hoc comparisons of lexical diversity across CDS, ADS, and written texts using Tukey’s HSD test.

Group 1	Group 2	MD	<i>p</i>	95% CI
ADS	CDS	−0.067	< .001	[−0.072, −0.061]
ADS	Written	0.074	< .001	[0.066, 0.082]
CDS	Written	0.141	< .001	[0.133, 0.148]

Table S4: Normalized lexical diversity values across registers. Lexical diversity was measured using MATTR and voc-D, computed for all words and for metaphorical words only.

Register	MATTR (all)	voc-D (all)	MATTR (metaphors)	voc-D (metaphors)
Child speech (CS)	0.399	0.284	0.338	0.133
Child-directed speech (CDS)	0.478	0.310	0.394	0.131
Spoken language	0.528	0.445	0.535	0.374
Written language	0.578	0.791	0.666	0.908

We also analyzed the most frequent word types annotated as metaphors, relative to the total number of metaphorical words in each corpus. The 20 most frequent metaphorical word types account for 71% in CDS, 51.5% in the spoken genre, 40% in the written register, and 73.4% of all metaphors in child speech. This distribution again shows that CDS is more repetitive, even in its use of non-literal vocabulary.

A mixed-effects model was fitted to examine whether lexical diversity in child speech increases with age, and the results are summarized in Table S5. An identical mixed-effects model was fitted to examine the relationship between lexical diversity in CDS and age, with random intercepts accounting for variability across the 14 caregiver–child dyads (Table S6). LD increased modestly with age in both CDS and CS. Although the fixed-effect estimate for age is statistically significant, the effect size is rather small. We therefore interpret this result with caution and view it as suggestive rather than conclusive evidence for age-related growth in lexical diversity during this developmental window.

These supplementary analyses show that child-directed speech (CDS) is characterized by high lexical redundancy and lower diversity than adult-directed registers. CDS thus relies on a relatively restricted vocabulary, both overall and within metaphorically used language. At the same time, lexical diversity in CDS and in child speech increased modestly with child age, although the estimated developmental effects were small. Lower lexical diversity in CDS aligns with classic and contemporary accounts of caregiver input as repetitive and lexically constrained (Newport et al., 1977; Snow, 1972; Tal & Arnon, 2024).

Importantly, reduced diversity should not be interpreted as impoverished input. Repetition can facilitate segmentation, memory consolidation, and stable form–meaning mapping,

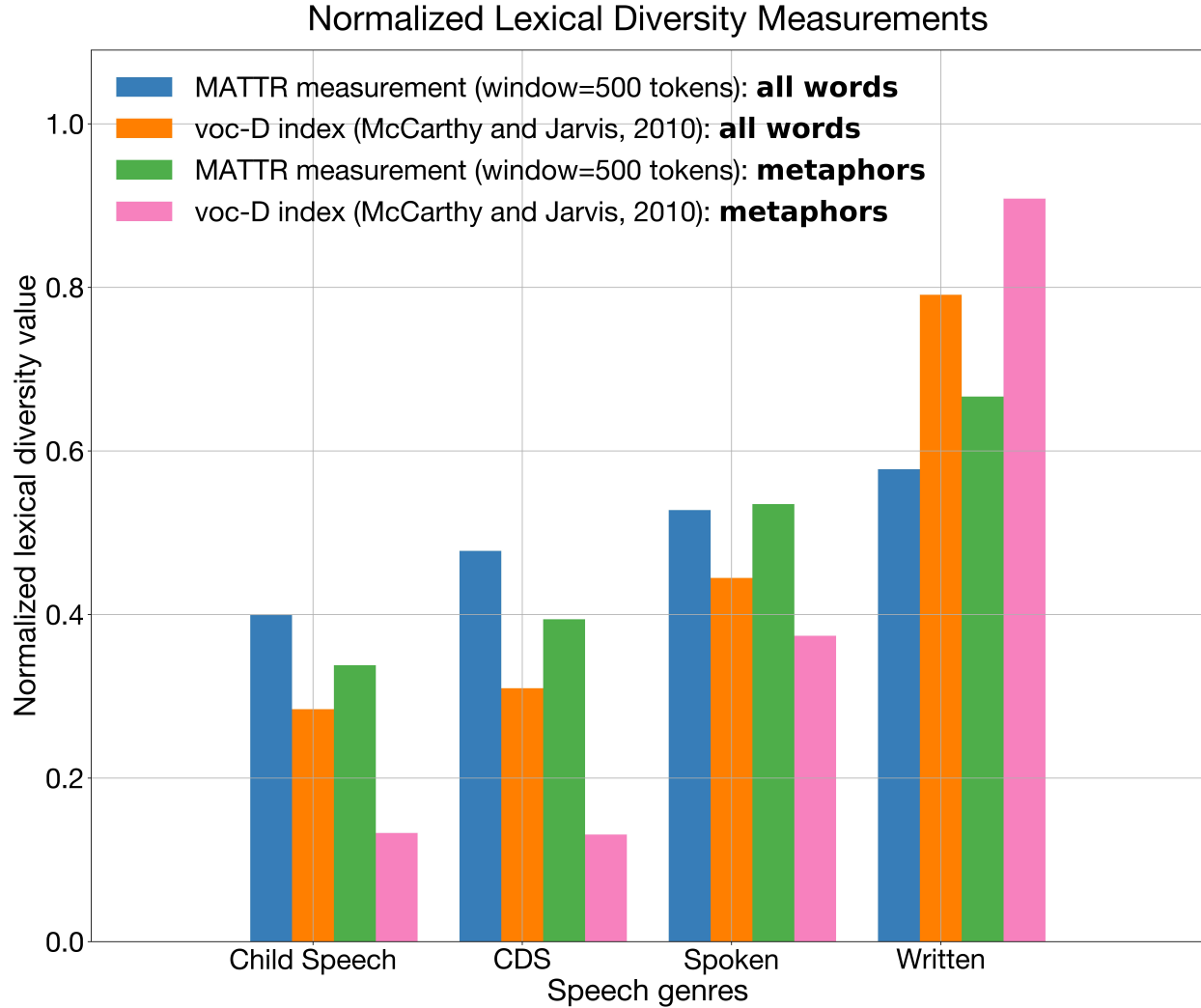


Figure S1: The normalized lexical diversity measurements for different registers. The output values represent both voc-D and MATTR calculations. The data were collected for all words in the analyzed corpora as well as metaphorical words only.

particularly in early learning contexts in which the same lexical items are repeatedly anchored to shared perceptual situations (Goldfield, 1999; Hills et al., 2009; Schwab & Lew-Williams, 2016). Within metaphorical language specifically, reduced lexical diversity suggests that caregivers tend to rely on a narrower inventory of figurative expressions. Such concentration may make figurative patterns easier to learn through repeated exposure, consistent with usage-based accounts in which entrenched constructions and recurrent mappings support abstraction and semantic extension (Bowdle & Gentner, 2005; Brochhagen & Boleda, 2023; Cameron-Faulkner et al., 2003).

The small but reliable increase in lexical diversity with child age is consistent with a gradual relaxation of redundancy as children’s processing capacities and lexical knowledge expand. However, the modest magnitude of this change suggests that CDS retains a strongly repetitive signature within the sampled developmental window, even as caregivers introduce

Table S5: Mixed-effects model results predicting lexical diversity in child speech (*mattr_CS*) from age.

Parameter	Est.	SE	<i>p</i>	95% CI
Intercept	0.544	0.012	< .001	[0.521, 0.567]
Age	0.041	0.002	< .001	[0.038, 0.044]
Group Var	0.002	0.020		

Table S6: Mixed-effects model results predicting lexical diversity in CDS (*mattr_CDS*) from age.

Parameter	Est.	SE	<i>p</i>	95% CI
Intercept	0.679	0.006	< .001	[0.666, 0.691]
Age	0.014	0.001	< .001	[0.013, 0.016]
Group Var	0.001	0.014		

somewhat more varied vocabulary. This pattern may reflect a functional trade-off: repetition continues to support learning while children progressively become able to benefit from richer lexical input. Crucially, the simplicity of early input involves not only how often words are repeated, but also what kinds of meanings these repeated words encode.

D. Metaphor Analysis Including the Written Register

For comparative purposes, we used a size-matched sample of written language from the British National Corpus (BNC), normalized to three million words (BNC Consortium, 2007; You et al., 2021). The BNC is a monolingual British English corpus comprising approximately 100 million words across spoken and written registers. The written component spans a range of genres, including newspaper articles, academic texts, and books (Burnard, 2007; McEnery & Hardie, 2011).

For transparency, we report here the supplementary analysis of metaphoricity across three registers: CDS, ADS, and written language. Consistent with the main analyses, CDS showed the lowest proportion of metaphorical words per speech or text sample. The mean percentages for each register are presented in Table S7 and visualized in Figure S2.

The one-way ANOVAs revealed significant differences in the percentages of metaphorical words among the three registers across both window sizes and both token- and type-based measures (Table S8). Across all analyses, CDS showed lower metaphorical-word percentages than both ADS and written language.

For the Window size 3 token-level analysis, Tukey’s HSD comparisons confirmed significant pairwise differences between all registers (Table S9). CDS showed lower metaphorical-token percentages than both ADS and written language, while ADS also showed lower metaphorical-

Table S7: Percentage of metaphorical words per speech or text sample relative to the total number of words across registers.

Register	Metaphorical words			
	Window size 100		Window size 3	
	Tokens	Types	Tokens	Types
CDS	5.77%	10.46%	5.55%	9.79%
ADS	11.86%	19.18%	8.98%	14.59%
Written	14.01%	21.95%	10.93%	16.05%

Table S8: ANOVA results for differences in the percentages of metaphorical words across CDS, ADS, and written language.

Statistic	Window size 100		Window size 3	
	Tokens	Types	Tokens	Types
$F(2, 980)$	1307.228	902.701	923.286	615.850
p	< .001	< .001	< .001	< .001

token percentages than written language.

E. Predictors of Children’s Metaphor Production and Vocabulary Growth

We used the Manchester corpus to examine supplementary relationships between caregiver input and child language production (Lieven et al., 2009; Theakston et al., 2001). These analyses extend the main manuscript descriptively by asking whether variation in children’s metaphorical word use and vocabulary growth is associated with properties of child-directed speech (CDS) within caregiver–child dyads.

To examine children’s metaphor production, we fitted mixed-effects models predicting the percentage of metaphorical word types in child speech and in CDS from session-level variables, including the percentage of metaphorical expressions in child speech and CDS, mean length of utterance (MLU), lexical diversity (MATTR), CDS concreteness, and the number of new word types produced by the child. Random intercepts were included for the 14 caregiver–child dyads. Random slopes were not included because of the small number of dyads and the risk of overparameterization.

We also fitted a mixed-effects model predicting the number of new word types produced in child speech from properties of CDS, including the percentage of metaphorical word types,

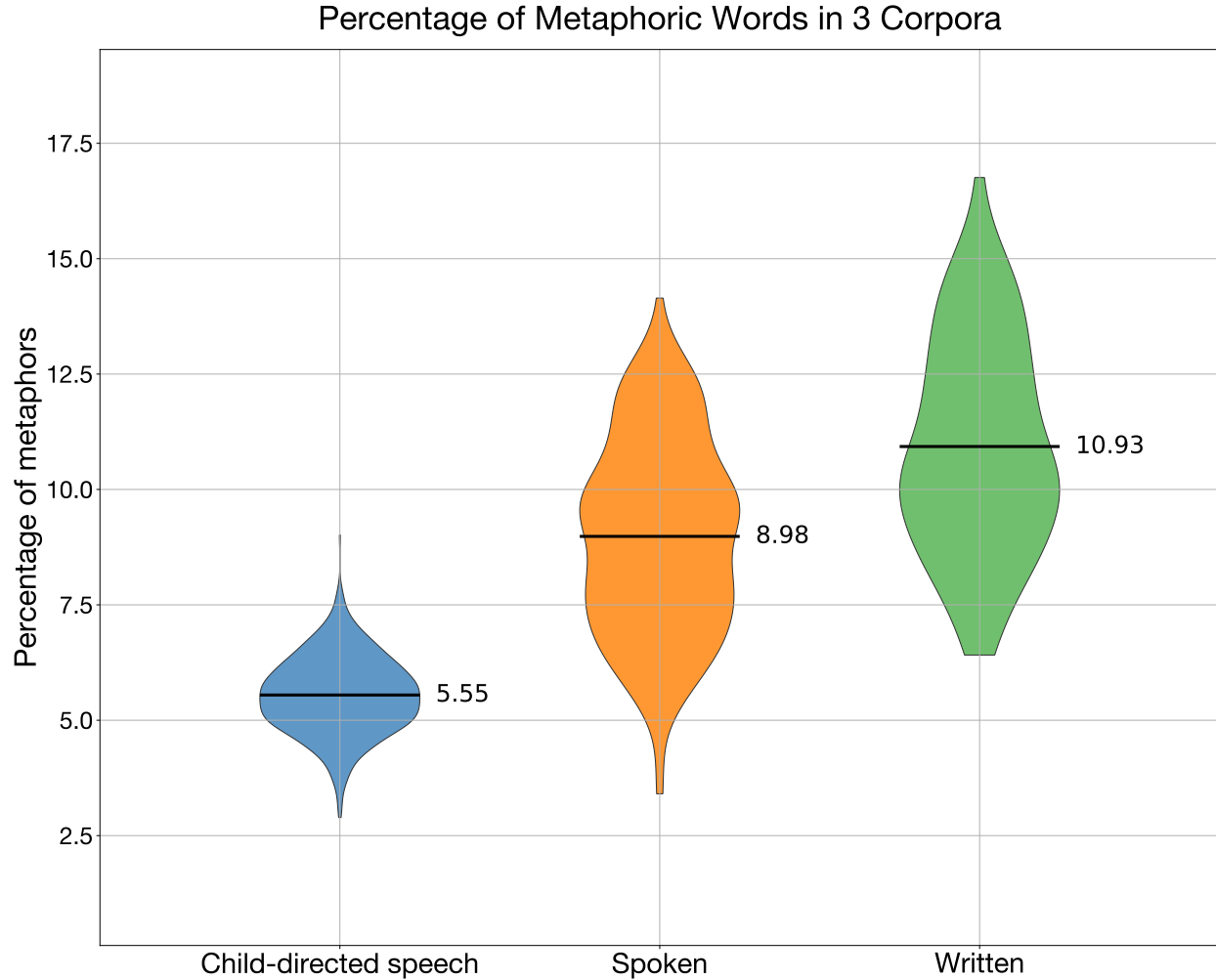


Figure S2: Violin plots of the percentages of metaphorical tokens per speech or text sample in the CDS, ADS, and written corpora. MelBERT computations were based on a window size of 3 tokens. Whiskers show the full range of the data distribution. Black solid lines indicate the means.

MLU, lexical diversity, and mean concreteness.

The mixed-effects model predicting the percentage of metaphorical word types in child speech (CS) revealed several findings. The results indicated that the percentage of metaphorical word types in CDS ($\beta = 0.397$, $p < .001$) was significantly associated with metaphorical word types in child speech. In addition, the mean length of utterance (MLU) in child speech ($\beta = 0.901$, $p < .001$) was significantly associated with the dependent variable. Other predictors, including the concreteness of CDS ($\beta = 0.733$, $p = 0.700$) and the number of new word types produced by children ($\beta = 0.003$, $p = 0.360$), did not show significant effects (Table S10).

The mixed-effects model predicting the proportion of metaphorical word types in CDS yielded several significant findings. The percentage of metaphorical types in child speech ($\beta = 0.158$, $p < .001$), mean length of utterance in CDS ($\beta = 0.434$, $p < .001$), and

Table S9: Post hoc comparisons of metaphorical-token percentages across registers using Tukey’s HSD test for the Window size 3 analysis.

G1	G2	MD	<i>p</i>	LB	UB
ADS	CDS	−3.47	< .001	−3.72	−3.21
ADS	Written	1.91	< .001	1.51	2.31
CDS	Written	5.38	< .001	5.01	5.74

Note. G1 = Group 1; G2 = Group 2; MD = mean difference, calculated as G2 minus G1; LB = lower bound; UB = upper bound.

concreteness of CDS ($\beta = -3.945$, $p = 0.001$) were significantly associated with the dependent variable. Other predictors, such as lexical diversity ($\beta = -1.995$, $p = 0.425$) and mean length of utterance in child speech ($\beta = -0.143$, $p = 0.302$), did not show significant effects (Table S11).

Finally, the mixed-effects model examining the relationship between features of CDS and the number of new word types produced by children revealed several significant effects. Mean length of utterance (MLU) in CDS ($\beta = -5.499$, $p = 0.033$) and lexical diversity (MATTR) in CDS ($\beta = -208.317$, $p < .001$) were both negatively associated with the number of new word types produced. In contrast, the concreteness of CDS ($\beta = 123.560$, $p < .001$) showed a significant positive effect. The percentage of metaphorical word types in CDS did not significantly predict new vocabulary growth ($\beta = -1.068$, $p = 0.225$) (Table S12).

These supplementary analyses examined associations between properties of caregiver input and child language production. In addition, the production of new word types was statistically associated with several properties of CDS, including concreteness, lexical diversity, and MLU in the reported models.

The positive association between metaphoricity in CDS and children’s metaphor production supports an input-sensitive account of figurative development. Children appear more likely to produce metaphors when their caregivers use more metaphorical language, suggesting that figurative usage is at least partly learned through exposure to and reuse of attested patterns. This interpretation aligns with accounts in which children acquire figurative meaning through participation in interaction and repeated encounters with conventionalized metaphorical expressions. The additional contribution of child MLU further indicates that metaphor production is intertwined with broader linguistic development: as children’s utterances become longer and structurally more complex, they gain greater capacity to express relational meanings and to combine lexical items in ways that support figurative extension.

The associations between CDS features and children’s production of new word types further suggest that input richness is multidimensional. Concreteness may facilitate early acquisition by supporting perceptual grounding and more stable inferential mappings between words and referents. At the same time, lexical diversity and grammatical complexity do not necessarily promote vocabulary growth uniformly within the sampled developmental window.

Table S10: Mixed-effects model results predicting the percentage of metaphorical word types in child speech (CS).

Parameter	Estimate	SE	<i>z</i>	<i>p</i>
Intercept	−2.318	6.590	−0.352	0.725
Metaphors in CDS (%)	0.397	0.063	6.299	< .001
MLU in CDS	−0.231	0.191	−1.209	0.227
MATTR in CDS	6.818	4.252	1.603	0.109
Concreteness of CDS	0.733	1.900	0.386	0.700
MLU in CS	0.901	0.227	3.966	< .001
MATTR in CS	−1.839	1.728	−1.064	0.287
New word types in CS	0.003	0.003	0.916	0.360
Group variance	0.768	0.210		

Depending on the child’s stage of development, more diverse or syntactically complex input may trade off against repeated, highly learnable mappings. Taken together, these findings are consistent with a developmental shift in which early learning benefits from grounded and redundant input, while later learning increasingly benefits from informational richness and structural variety.

Where direct caregiver interaction occurs, these supplementary findings are consistent with an input-sensitive, usage-based account of metaphor acquisition and lexical creativity. Rather than pointing to sudden, internally driven conceptual leaps alone, the observed associations suggest that children exposed to more metaphorical language from their primary caregivers also tend to produce more nonliteral expressions themselves. This direct distribution tracking is highly consistent with usage-based theories of language development, which emphasize that abstract, creative, and figurative constructions emerge gradually through the entrenchment and structural schema-building of concrete linguistic tokens encountered within the child’s actual input environment (Dąbrowska & Lieven, 2005; Hartmann et al., 2021; Lieven et al., 2003; Tomasello, 2003).

F. Predicting Metaphoricity Rate of a Word Type from its Relative Frequency, Dictionary Frequency, and Concreteness Score

As part of our analyses, we examined variables associated with individual word types. Specifically, we analyzed the relative frequency of each word in our corpora, dictionary-based word frequency, the proportion of metaphorical usage in each corpus (i.e., the frequency of metaphorical annotations relative to the total occurrences of the word), and concreteness

Table S11: Mixed-effects model results predicting the proportion of metaphorical word types in CDS.

Parameter	Estimate	SE	<i>z</i>	<i>p</i>
Intercept	18.978	3.973	4.777	< .001
Metaphors in CS (%)	0.158	0.024	6.493	< .001
MLU in CDS	0.434	0.109	3.973	< .001
MATTR in CDS	−1.995	2.503	−0.797	0.425
Concreteness of CDS	−3.945	1.149	−3.433	0.001
MLU in CS	−0.143	0.138	−1.032	0.302
MATTR in CS	1.221	1.068	1.144	0.253
New word types in CS	−0.003	0.002	−1.605	0.109
Group variance	0.103	0.053		

Table S12: Mixed-effects model results predicting the number of new word types produced by children from features of CDS.

Predictor	Estimate	SE	<i>z</i>	<i>p</i>
Intercept	−123.946	91.940	−1.348	0.178
Metaphors in CDS (%)	−1.068	0.880	−1.213	0.225
MLU in CDS	−5.499	2.582	−2.130	0.033
MATTR in CDS	−208.317	52.081	−4.000	< .001
Concreteness of CDS	123.560	26.163	4.723	< .001

ratings derived from both continuous and categorical datasets. Dictionary word frequency was based on the SubtlexUS database by Brysbaert and New (Brysbaert & New, 2009), compiled from movie and TV series subtitles (51 million words). We utilized frequency values reported as occurrences per million words, focusing on a subset of 60,384 words with a frequency greater than one.

Our dataset included all word types present in our corpora for which concreteness ratings and dictionary frequency data were available. This yielded 5,918 words in the child-directed speech corpus, 13,680 words in the adult-directed speech corpus, and 19,077 words in the written corpus. For each word type, we incorporated its relative frequency, dictionary frequency, and metaphoricity proportion. After normalizing variable values, we applied a mixed-effects model to predict the metaphoricity of word types based on their relative frequency in the corpus, dictionary frequency (Brysbaert & New, 2009), and concreteness ratings as fixed effects, with a random intercept for each word type.

The results indicate that concreteness rating was the only predictor that was consistently associated with word-type metaphoricity across all corpora. This pattern was consistent for both continuous and categorical concreteness ratings. Specifically, Table S13 summarizes the fixed-effects estimates for the concreteness rating across the three evaluated language registers.

Table S13: Fixed-Effects Estimates for Concreteness Rating Predicting Word-Type Metaphoricity across Corpus Registers.

Register	<i>N</i> (Words)	β	<i>t</i>	<i>p</i>
Child-directed speech (CDS)	5918	−0.010	−7.253	< .001
Adult-directed speech (ADS)	13 680	−0.005	−6.846	< .001
Written language	19 077	−0.002	−8.102	< .001

The negative coefficient suggests that words with higher concreteness ratings are less likely to be used metaphorically. Figure S3 illustrates this negative association between continuous concreteness scores and word-type metaphoricity. In other words, the proportion of metaphorical uses decreases systematically for more concrete words.

Figure S4 juxtaposes categorical rating distributions, showing that the mean metaphoricity rate of word types is lower for concrete words than for abstract words across all genres.

These findings align with the conceptual framework proposed by Lakoff and Johnson (Lakoff & Johnson, 1980), which posits that metaphorical expressions transfer knowledge from concrete source domains to abstract target domains, often involving a shift from a more concrete source-domain meaning to a more abstract contextual meaning. For a word to be metaphorical in a given context, it must possess both a concrete and an abstract sense, with the abstract sense activated in that context (Maudslay & Cotterell, 2020). Since concreteness ratings typically reflect the averaged or dominant sense of a word, metaphorical usage is more prevalent among words whose dominant or averaged meanings are relatively abstract.

Conversely, words with the highest concreteness ratings are often evaluated based on their original, source-domain meanings or lack an abstract sense altogether, making them less likely to appear in metaphorical contexts. This effect is particularly pronounced in child-directed speech, where concrete words are predominantly used in literal contexts, ensuring the availability of source-domain meanings during early language exposure.

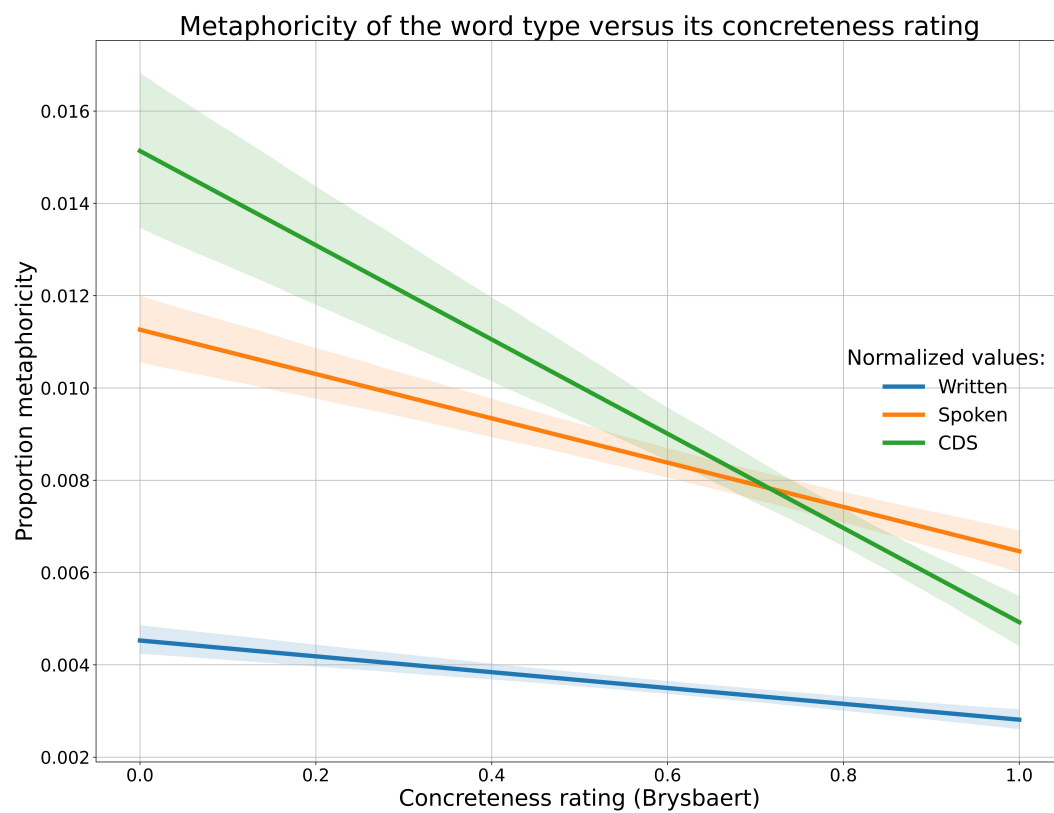


Figure S3: Linear relationship between normalized concreteness scores and metaphoricity rates of word types across registers.

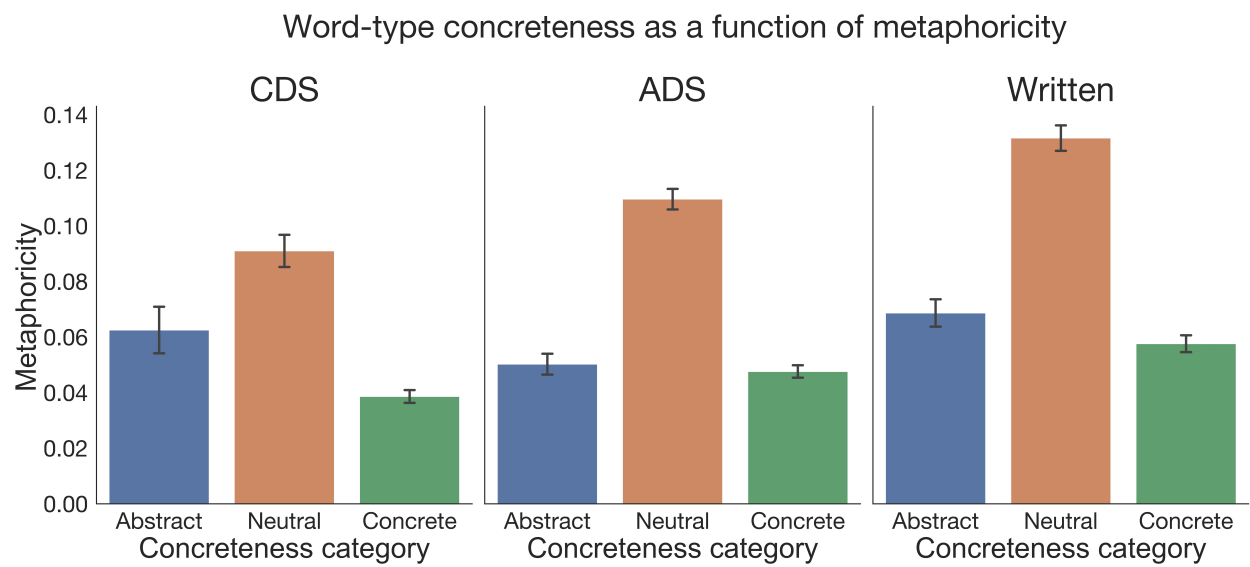


Figure S4: Word-type concreteness as a function of metaphoricity across registers. Bars represent mean metaphoricity by concreteness category; error bars indicate 95% confidence intervals.

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